



BEHAVIOURAL OPERATIONS

QUESTION: 1

1a: According to expected utility theory, what is the optimal offer of the proposer?

Please state your whole reasoning.

The ultimatum game pertains to a gift of £10 endowment in this experiment with the ability for the proposer to give any amount from £1 to £9 to the responder. The responder can accept the offer and receive £x (the amount offered), while the proposer receives £ (10 – x) or reject it, in which case both of them receive £0.50. Such consequences are common knowledge to both players.

Expected Utility Theory (EUT) assumes that agents are rational and seek to maximize their own utility relatively, typically equated with monetary payoff in experimental economics. From this, it is expected that the responder will always have to accept those reasonable offers above £0.50 since not accepting it will yield only £0.50.

Since offers are restricted to whole pounds, it follows that the lowest acceptable amount should be £1 which satisfies the responder's condition ($£1 > £0.50$). The rational proposer, therefore, finds it most beneficial to offer £1 while keeping £9 for himself because he would not need anyone else to accept this. The following strategy maximises the proposer's payoff while still ensuring the responder's acceptance.

Optimal offer = £1

However, this result is not unexpected, but it is purely economic rationality and the assumption that responders will take anything greater than their rejection fallback. EUT does not account for considerations of fairness, emotion, or social norms focusing solely in self-interested maximisation of payoffs.

We analyse the ultimatum game under the assumption that all players are perfectly

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- rational and strive to maximize expected utility equating to monetary value in this context.
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Let:

- $x \in \{1, 2, \dots, 9\}$ the amount offered to the responder
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The proposer receives $10 - x$ if the offer is accepted

Both proposer and responder receive 0.5 if the offer is rejected

Let U_P and U_R denote the utilities of the proposer and responder respectively.

The responder will accept the offer if and only if:

$$U_R = x > 0.5$$

This gives us the **acceptance constraint**:

$$x > 0.5 \Rightarrow x \geq 1$$

Since only whole pounds can be offered, the **lowest acceptable offer is $x = 1$** The

proposer's objective is to:

$$\max_{x \in \{1, 2, \dots, 9\}} U_P = 10 - x$$

Subject to:

$$x \geq 1$$

The proposer's utility U_P is **strictly decreasing** in x , so the optimal solution is:

$$x^* = 1 \Rightarrow U_P = 10 - 1 = 9$$

This yields:

- Proposer receives £9
- Responder receives £1
- The responder accepts since $1 > 0.5$

Conclusion (EUT Prediction):

Under Expected Utility Theory, the proposer should offer £1 to maximise their utility, satisfying the constraint that the responder receives more than £0.50.

Question 1b: Based on data from the ultimatum game lab experiments discussed in class, do you think that the prediction of expected utility theory in (a) would be accurate? Why or why not?

While EUT offers a prediction that is logically consistent, observations made in real life deviate significantly from its assumptions and conclusions from laboratory results coming from ultimatum game experiments.

As the following empirical results demonstrate:

- Offers below £2-£3 are often rejected frequently even though this leads to lower monetary outcomes for the respondents.

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Equal-near equal splits such as (£5/£5 or £4/£6) have been the most possible offers made and

- accepted.

Proposers view low offers as being rejected most of the time and adjust favouring making generous offers.

These behaviours cannot show that responders are not self-interested; rather, they point out that decisions are muddled up with considerations of fairness, with emotional impulses (anger, spite), or wanting to punish what they see as greed-inclined behaviour even at a personal cost. These findings are based on various theories of behavioural economics, including ones incorporating social preferences and psychological utility.

Therefore, this prediction derived from EUT - an offer of £1 by a rational proposer would always be accepted - is patently inaccurate in reality. Actual behaviour is influenced by a much more complicated and embedded social decision process than that assumed by EUT.

Although EUT suggests that the proposer should put forward an offer of £1, whereas experiments in the laboratory demonstrate that offers of this value are frequently rejected by the receiver.

Here's why the theoretical prediction fails in real-life behaviour:

Empirical Constraint:

Responders typically reject offers below £2–£3, even though this results in a lower payoff for them.

We can now redefine the behavioural acceptance constraint:

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$$x \geq x_{\text{fair}} \quad \text{where } x_{\text{fair}} \approx 2 \text{ to } 3$$

This originates from:

- **Fairness concerns** – people care about whether the split is fair.
- **Negative utility** - Witnessing someone receiving more than yourself (and sometimes even less) can set off discomfort in a person and reduce their satisfaction - or their "utility".
- **Social preference** for division equally.

$$U_R^{\text{beh}} = x - \alpha(10 - 2x)$$

Where:

- $\alpha > 0$ is a weight on inequity aversion
- $10 - 2x$ is the inequality in the split
- The responder may reject offers which implies that $U_R^{\text{beh}} < 0.5$

Behavioural Conclusion:

Because utility functions are motivated by fairness, a proposed offering between £3 and £5 occasions a considerable chance for rejection when the recipient deems the offer too low. Thus, the outcome of the games will be determined not specifically by utility maximising, but more so by social norms, emotions, and considerations of fairness.

Final Summary:

Factors	EUT Prediction	Behavioural Data
Optimal Offer	£1	£3–£5
Acceptance Threshold	$x > 0.5$	$x \gtrsim 2-3$
Proposer Strategy	Maximise $10-x$	Balance payoff including fairness
Model Limitation Behaviour	Excludes social preferences	Captures real-world behaviour

QUESTION: 2

Critically evaluate the extent to which theories and research findings from the behavioural sciences can contribute to better decision making in operational contexts.

Introduction

Operational decision-making has been based on models with rational utilitymaximizing agents however these have been challenged by behavioural science-from psychology,

behavioural economics, and cognitive science-as an arguably more realworld lens that highlights how the individual deviate from rationality in systematic ways. In the report, its critical evaluation brings in behavioural theories and findings as further enhancements to decision-making operational contexts beyond benefits, limitations, and applicability.

1. Bounded Rationality and Heuristics

Theory:

According to Herbert Simon who introduced the concept of bounded rationality, human beings are cognitively bounded by limitations in their ability to process complex information. As a result, they employ heuristics (mental shortcuts) instead of exhaustive optimisation.

Operational Application:

For instance, a warehouse manager might reorder stock once inventories "look low" instead of running a detailed EOQ (Economic Order Quantity) calculation. In manufacturing scheduling, workforce allocation, or inventory control, managers often resort to the use of "rules of thumb," rather than solving complex models.

Critical Evaluation:

Heuristics can be effective under time constraint but also lead to systematic errors (for example, overconfidence or availability bias). Understanding of how people think naturally leads to better-designed decision support systems and more relevant training.

2. Prospect Theory and Risk Behaviour

Theory:

Prospect Theory, developed by Kahneman and Tversky, shows that people:
Are loss averse - Losses incite more intense feelings than commensurate gains
evoke in positive emotions.

Evaluate outcomes relative to a reference point - People compare and evaluate
the outcomes of the events according to those events' reference points.

Probability Weighting - People tend to overreact to small probabilities and
underreact to big ones.

Operational Application:

They would not even be aware of the risk-adjusted return of operators in maintenance planning or risk management. For instance, operators could react too strongly to a lastminute failure of the machine (availability bias) or would avoid a new process because of the perceived risk of losing something. Reframing alternatives (for example, avoiding loss rather than achieving gain) would work in influencing their perception of risk.

Critical Evaluation:

Although useful, the absence of a universal formula to integrate Prospect Theory in optimization models makes it difficult to apply the theory because one has to account for an individual's reference point or utility curves as these may differ from person to person.

3. Social Norms and Peer Effects

Theory:

The way behaviour is influenced by what others are doing is called descriptive norms, whereas the expectation others may have for that behaviour is an example of injunctive norms.

Operational Application:

In lean operations or safety management, metrics such as "90% of teams achieved zerodeflect days in this week" use peer comparison for performance enhancement purposes. **Critical Assessment:**

When nudges are attributed to social norms, their effectiveness varies according to the underlying culture of the organization. If poorly implemented, they can backfire (for instance, if the perceived social norm is one of low performance). Norm fatigue can also cause deterioration in the effectiveness of such norms over the long term.

4. System Thinking and Mental Models

Theory:

Understanding dynamic systems is generally difficult for individuals, especially those entailing delays, feedback, and accumulations in the System Dynamics paradigm. The “stock-flow failure” is a cognitive error which refers to the misinterpretation that people have of how inventories gain or deplete.

Operational Application:

Behavioural science would suggest that to create mental models in the settings of supply chain or project management, visual simulations or other interactive dashboards would be used.

Critical Evaluation:

More so than good in contributing to understanding, these very instruments depend on user involvement at hands and the previous training of the facilitator. Some employees would not learn equally effectively through use of visual means.

5. Motivation and Incentive Design**Theory:**

According to behavioural science, money is merely a portion of what people seek; autonomy, mastery, and purpose are others (according to Deci & Ryan's Self-Determination Theory).

Operational Application:

When recognition, accountability, and skill-building shine far more brightly than cash rewards in quality circles or continuous improvement projects, their applications often lead to increased productivity and satisfaction. Empowering teams even nominal ownership over decision-making enhances productivity and satisfaction.

Critical Evaluation:

Behavioural incentives are contextual in nature. What runs well in one culture or organization may falter in another. Even worse, poorly designed incentive schemes may crowd out intrinsic motivation (crowding-out effect).

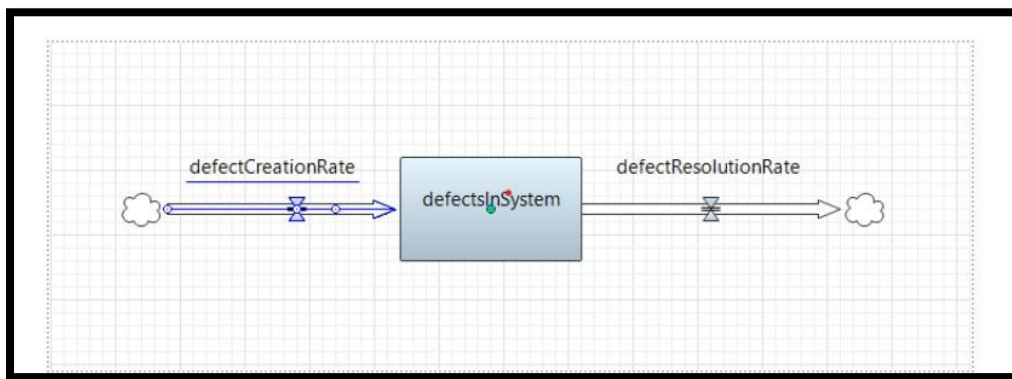
Conclusion

Behavioural science provides important insights on human factors, while such human factors are often disregarded by traditional operational research. In its realm of nudging

and designing customised incentives and decision aids, behavioural science applies a more realistic model and engages better and leads to improved outcomes. Yet, behavioural approaches require careful considerations about context, complexity, and communication in their applications. The most successful solutions are those blending behavioural realism with operational rigour.

QUESTION: 3

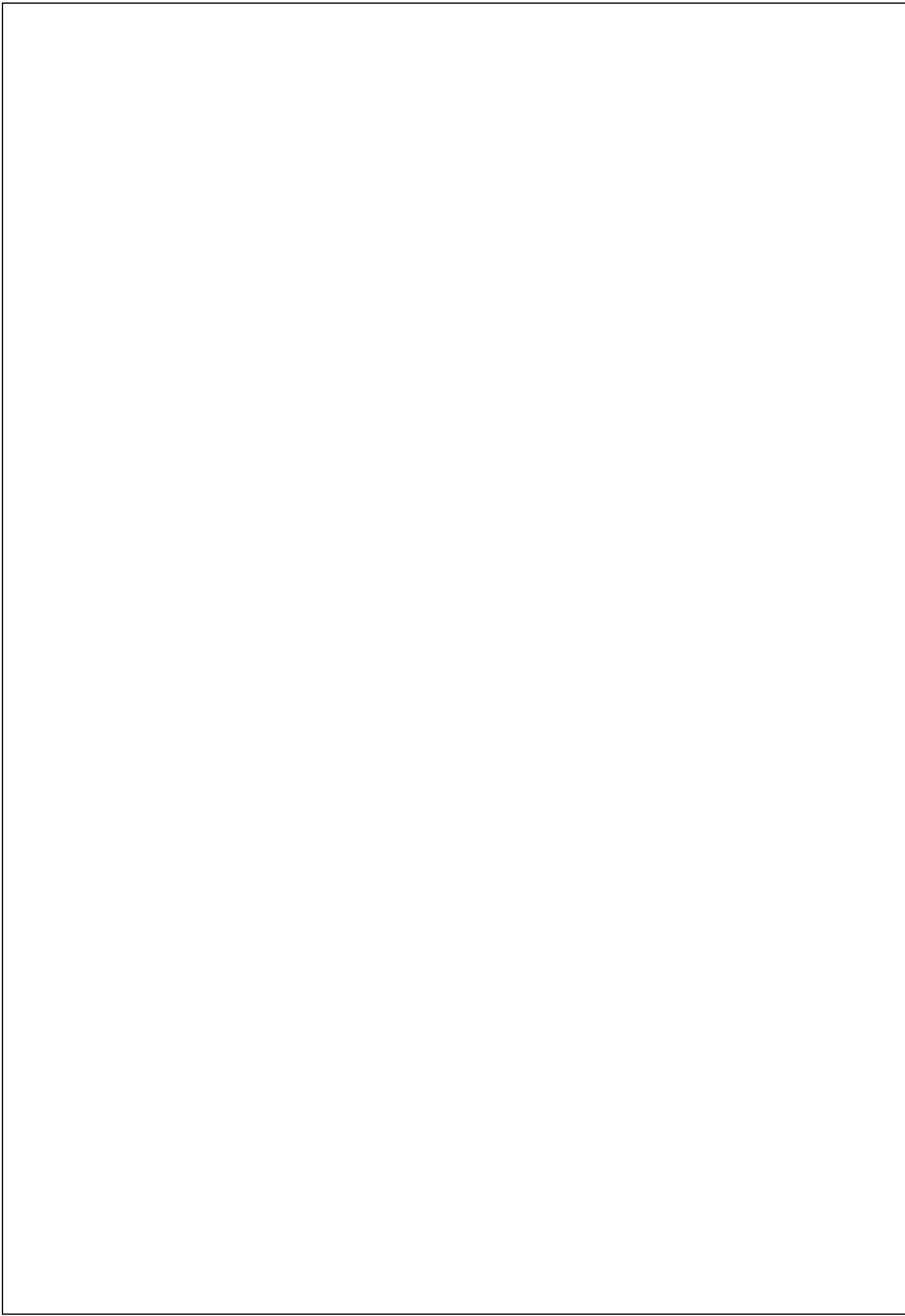
3a. Decide whether defects are stock or a flow and draw a stock flow structure representing the creation of defects and elimination of defects. Explain your reasoning.



A stock represents something which may increase or decrease over time in system dynamics. In the context of a factory, defects can be described as a stock because they accumulate when new ones are created and decrease when they have been fixed. The defect creation rate adds defects to the system and is the flow of defects as they are created. The exit flow is called defect resolution rate, and it is the rate of defects getting out of the system.

The model structure consists of:

- **Stock:** defectsInSystem - representing the present number of unsolved defects.
- **Inflow:** defectCreationRate - the speed with which new defects occur. This flow describes the number of new defects added to the system. It depends on factors such as:



Machine condition or wear-and-tear of the machine, the quality of raw materials, operator error or fatigue, process inefficiencies.

- **Outflow:** defectResolutionRate - the speed at which defects get fixed. Factors are - Availability of repairing resources or maintenance staff, Speed and effectiveness of quality control, Frequency of inspections or process audits.

Mathematical Representation (Stock Equation):

The change in the number of defects over time can be described using this differential equation:

$$\frac{d(\text{defectsInSystem})}{dt} = \text{defectCreationRate} - \text{defectResolutionRate}$$

Where:

$\frac{d(\text{defectsInSystem})}{dt}$ is the rate of change in the number of defects,

defectCreationRate is the rate of incoming defects,

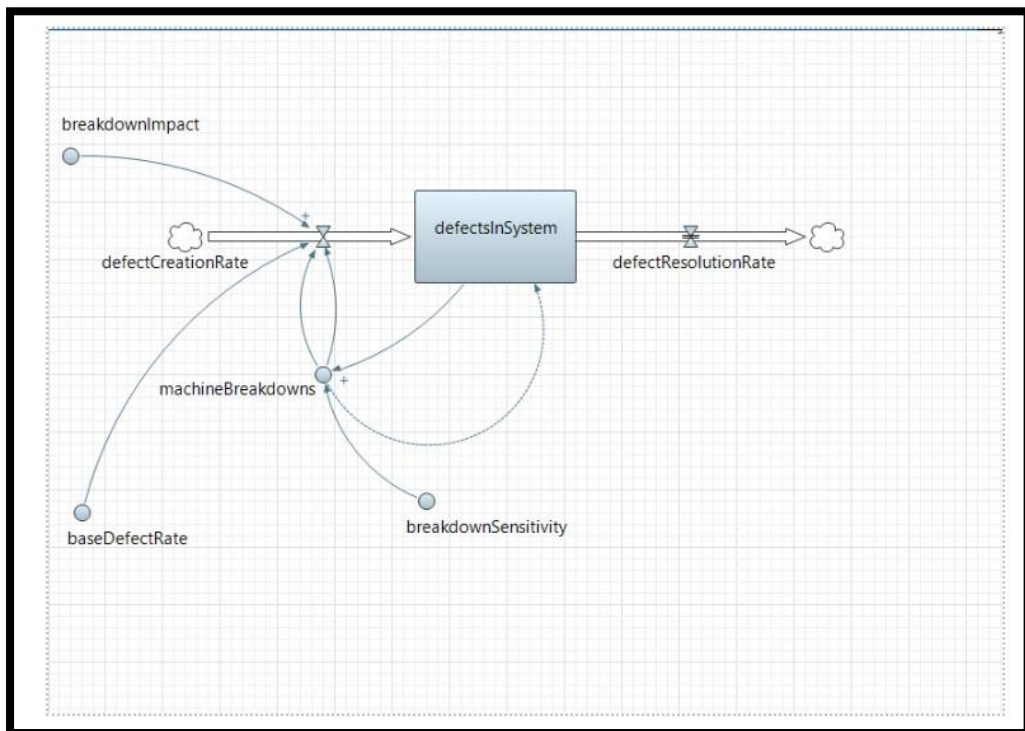
defectResolutionRate is the rate at which defects are resolved.

- This stock-flow structure proves beneficial since it gives us insight into how the number
- of defects processes in the course of time depending on the speed in which they are created and resolved. If the defects are created faster than they are resolved, they will
 - accumulate in the system, causing quality issues and operational delays.
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On a counter note, if the factory resolves defects faster than they are created, the number of defects will be reduced, thereby improving product quality and efficiency. The

management can use such defect modelling to simulate different scenarios and make informed decisions on resource allocation to stay in control of quality and avoid any overload on the system.

3b. Now consider how defects cause breakdowns which then cause further defects. Add a loop to show this in your stock flow diagram. Explain the loop, including its polarity. What behaviour would you expect over time? Consider different situations.



In the extended version of the stock and flow model, we look at the second-order effect of some existing defects contributing to the failure of the system, that is, breakdown of machines. The main idea is that with the increasing number of unresolved defects, the likelihood of machine breakdowns would increase. The breakdowns further interfere with the production process, giving rise to even more defects.

Now this cycle forms a reinforcing feedback loop, that is, a positive loop in which one part of the system strengthens over time. It shows how a small defect (few defects initially) can grow bigger if no action is taken to curb the development. A positive feedback loop, if not suppressed, might eventually cause instability or increased degradation of the system.

The usual in manufacturing and operations, where system stress increases and breakdowns multiply due to the inability to rectify quality problems, culminates in cascading failure.

Those components are:

- **Stock:** defectsInSystem - the sum of in-work unresolved or ongoing defects located in the system.
- **Inflow:** defectCreationRate - creation rate of new defects based on machine breakdowns.
- **Outflow:** defectResolutionRate - the rate at which defects are repaired and taken from the system.

Intermediate Variable: machineBreakdowns - which is an inverse function of defect levels, that is, more defects higher breakdowns.

Constants or Multipliers:

baseDefectRate: The natural rate of defect creation when there are no active breakdowns.

breakdownSensitivity: This defines to what extent the present defect level affects the frequency of breakdowns.

breakdownImpact: The extent of increase in defect creation due to breakdowns.

Mathematical Relationships:

1. **Machine Breakdowns** increase with defects:

$$\text{machineBreakdowns} = \text{defectsInSystem} \times \text{breakdownSensitivity}$$

2. **Defect Creation Rate** increases with breakdowns:

$$\text{defectCreationRate} = \text{baseDefectRate} + (\text{machineBreakdowns} \times \text{breakdownImpact})$$

3. **Stock Accumulation Equation:**

$$\frac{d(\text{defectsInSystem})}{dt} = \text{defectCreationRate} - \text{defectResolutionRate}$$

Loop Polarity and its Dynamics:

- Feedback loop (positive) reinforcement:

More defects → more breakdowns → more defects

- The reason for a positive polarity is that each cycle of the loop amplifies whatever minute change started the process.

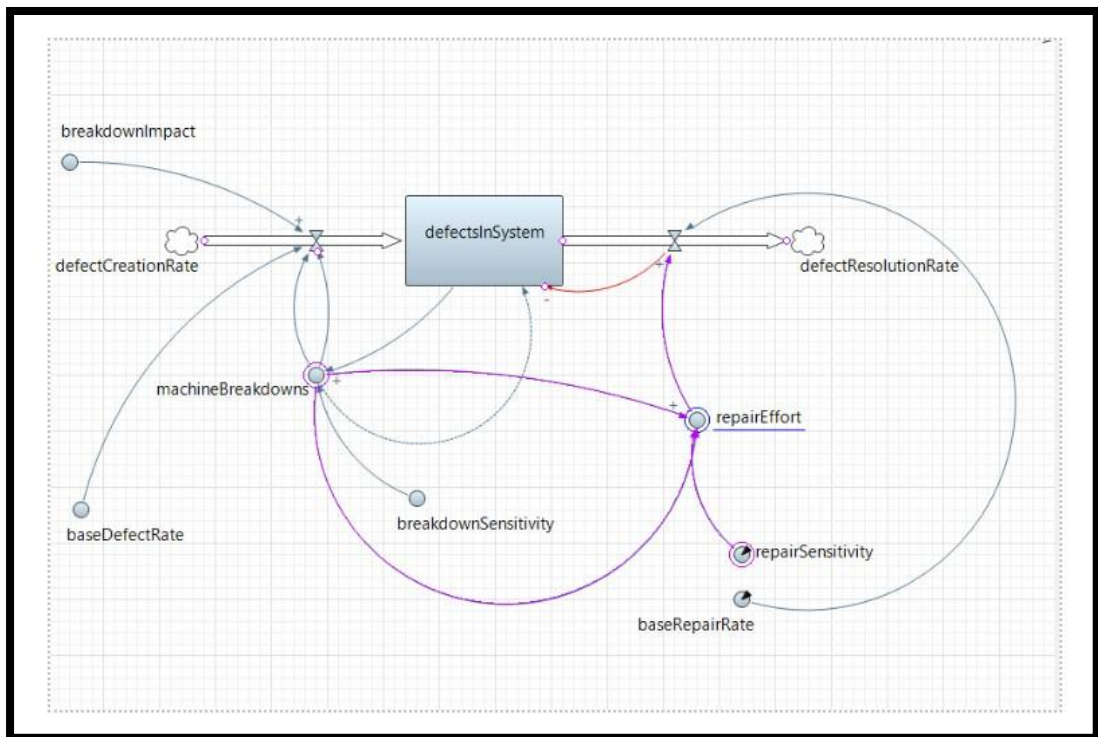
- Such loops unless balance is achieved (e.g., increased repairs, preventive maintenance) could lead to an exponential increase in defects.

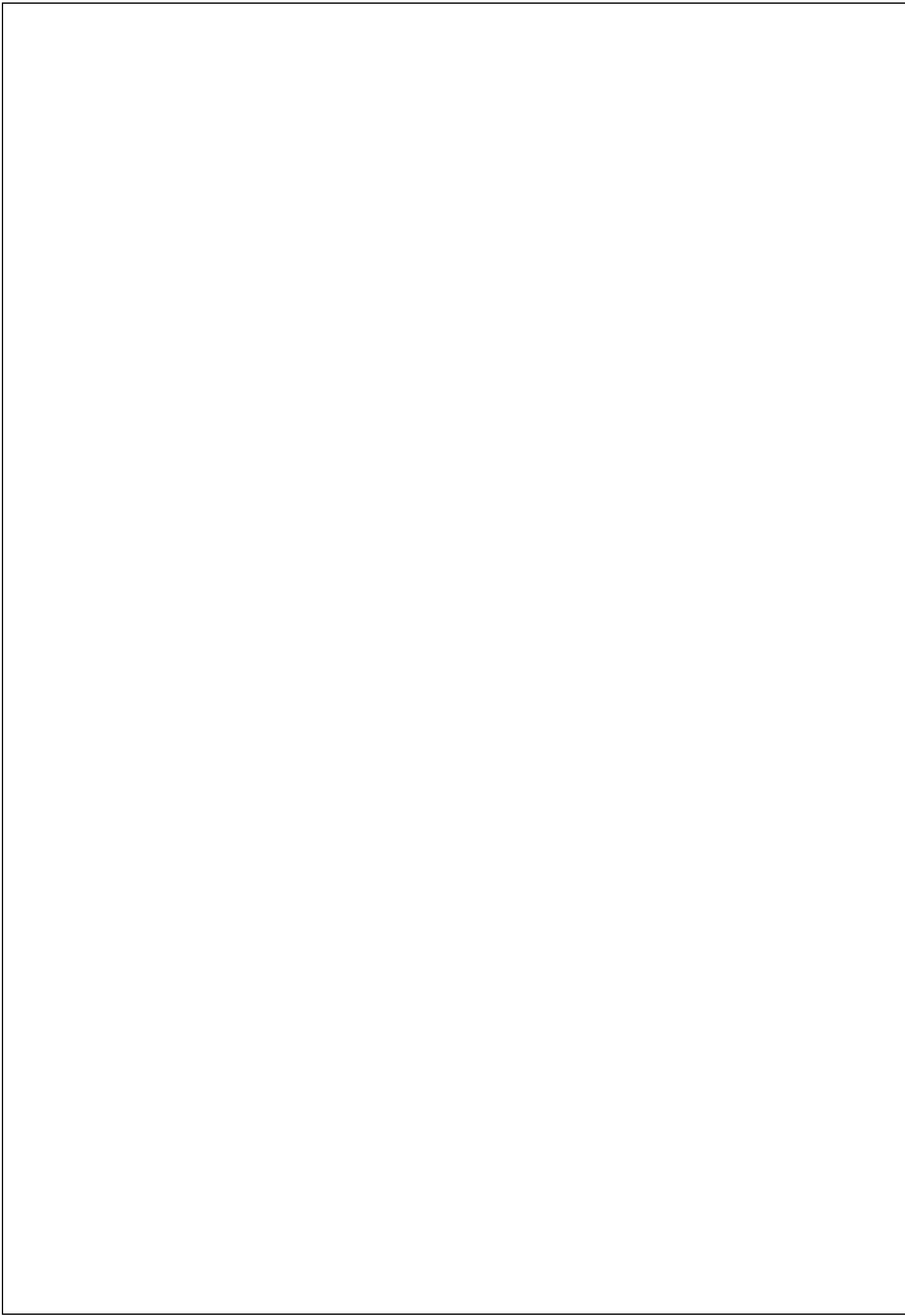
Expected Behaviour of the System over Time:

- With a small base defect rate and low feedback effect (breakdown sensitivity and impact), defect levels may remain low or stable.
- On the contrary, if the feedback is large, even a few defects can cause a lot of breakdowns and rapidly increase the total defect level.
- The system can enter a self-reinforcing spiral if a failure typically in a poorly maintained or overstressed manufacturing environment.

This part of the model emphasizes that prompt detection of defects and prevention of breakdowns are critical because a small defect can turn into a larger impasse if left unattended.

3c. Breakdowns would encourage repair efforts to eliminate the defects. Add structure to show this in your stock flow diagram. Explain the loop, including its polarity. What behaviour would you expect over time?





The diagram shows how defects in a production system, resulting in breakdowns, initiate repairs to restore control over the system. At the very centre is the state variable called `defectsInSystem`, which represents the total number of unresolved defects.

Defects are accumulating through the inflow `defectCreationRate`; the latter is influenced by `machineBreakdowns`. These breakdowns do not occur at constant rates: rather, they increase over time according to the increase in defects, governed by a variable called `breakdownSensitivity`. As breakdowns increase, they activate a proportional response in `repairEffort`, which accumulates with `repairSensitivity` and adds to the base repair rate (`baseRepairRate`). Increased repair effort is routed to increasing the `defectResolutionRate`, which is the outflow from the system and attempts to reduce defect counts over time.

The set of relationships thus forms a balancing (negative) feedback loop whereby greater defect numbers lead to more breakdowns, consequently triggering more repairs in an effort to reduce defects. This arrangement depicts a real-world dynamic where systems tend to scale resources to counter increasing failure.

Updated Components:

- **Stock:** `defectsInSystem` – the total number of unresolved defects.
- **Inflow:** `defectCreationRate` – increased by `machineBreakdowns`.
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Outflow: defectResolutionRate – now linked to repairEffort.

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Variables Introduced:

- **repairEffort**: the effort allocated to fixing defects, increases with breakdowns.
- **baseRepairRate**: the default repair activity.
- **repairSensitivity**: how much repair effort responds to breakdowns.

Equations:

1. Machine Breakdowns:

$$\frac{d(\text{defectsInSystem})}{dt} = \text{defectCreationRate} - \text{defectResolutionRate}$$

2. Repair Effort:

$$\text{repairEffort} = \text{baseRepairRate} + (\text{machineBreakdowns} \times \text{repairSensitivity})$$

3. Defect Resolution Rate:

$$\text{defectResolutionRate} = \text{repairEffort}$$

4. Stock Change Equation (same as before):

$$\frac{d(\text{defectsInSystem})}{dt} = \text{defectCreationRate} - \text{defectResolutionRate}$$

Loop Polarity:

This is a balancing (negative) feedback loop:

- More defects lead to more breakdowns, more repair efforts, faster resolution of

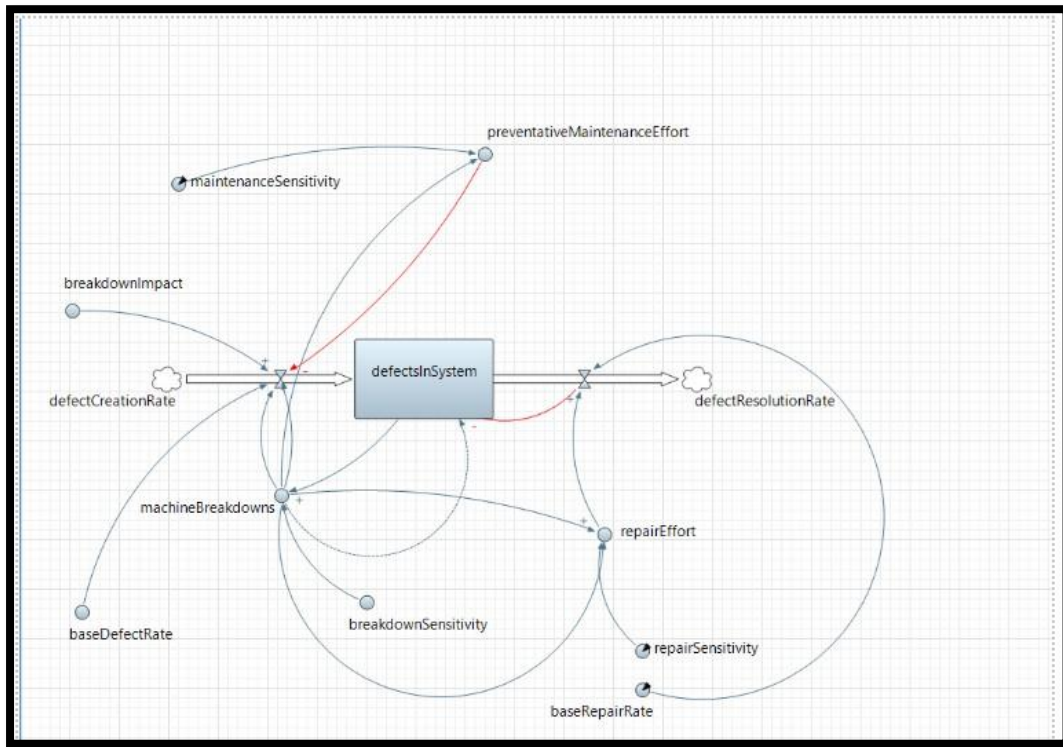
defects, and consequently fewer defects.

Expected Behaviour Over Time:

- Strong repair sensitivity means that this loop can recover the system to a controllable situation and decrease the total number of defects.
- If the repair response is weak or delayed, then the system may be suffering from rising defect levels.
- With reinforcing and balancing loops active simultaneously, oscillations or stabilized behaviour can be observed in the system, depending on the parameter values used.

This model represents a realistic scenario, in which, as the problems increase, the repair capacity is increasing, thus creating a self-correcting system that limits the escalation of defects if properly managed.

3d. Now imagine a new factory manager puts more emphasis on preventative maintenance as breakdowns increase. Add the required structure to your diagram and describe the loop and its polarity. What behaviour would you expect?



Explanation:

The diagram shows how the factory scheme dynamically reacts to increasing breakdowns by investing in preventive maintenance. At the heart of the schematic is a stock, `defectsInSystem`, which represents the total number of unresolved defects. Defects are added at the `defectCreationRate`, which increases with `machineBreakdowns`, a variable positively influenced by the current number of defects and modulated negatively by `breakdownSensitivity`.

The path brings an additional new element in this diagram. With increasing breakdowns, more efforts toward preventive maintenance are steered by, leading to defect facilities from a `maintenanceSensitivity` parameter. This effort decreases the `defectCreationRate` and prevents new defects from moving into the system.

Meanwhile, the model keeps the earlier structure to `repairEffort`-enhanced by breakdowns and `repairSensitivity`-increasing the `defectResolutionRate` to resolve the

standing defects. The addition of preventive maintenance compounds the balancing secondary feedback loop: not only is the system reacting to defects with repairs, it is now also addressing the root causes of those defects by reducing the inflow of defects. Such a setup strengthens the system as it brings balance between short-term repair solutions and long-term stability through prevention.

Model Structure Additions:

- **New variable:** preventativeMaintenanceEffort
- **New parameter:** maintenanceSensitivity
- **Effect:** As machineBreakdowns increase, preventativeMaintenanceEffort increases, which lowers the defectCreationRate.

Updated Relationships and Equations:

1. Preventative Maintenance Effort:

$$\text{preventativeMaintenanceEffort} = \text{machineBreakdowns} \times \text{maintenanceSensitivity}$$

2. Adjusted Defect Creation Rate:

$$\text{defectCreationRate} = \text{baseDefectRate} + (\text{machineBreakdowns} \times \text{breakdownImpact}) - \text{preventativeMaintenanceEffort}$$

3. Defect Stock Equation:

$$\frac{d(\text{defectsInSystem})}{dt} = \text{defectCreationRate} - \text{defectResolutionRate}$$

Feedback Loop Polarity:

This is a balancing (negative) loops:

More breakdowns → more preventative maintenance → lower defect creation rate → fewer future defects.

This loop prevents the accumulation of defects by reducing inflow rather than increasing outflow through repair.

Expected behaviour over time:

If the preventative maintenance response is strong and timely, the system will significantly decrease creation of future defects, subsequently stabilizing the entire system over time.

If the response is too weak or late, an increase in breakdowns will require repairs that are reactive to further lowering the system efficiency.

Thus, the combination of: reinforcing loop (defects → breakdowns →

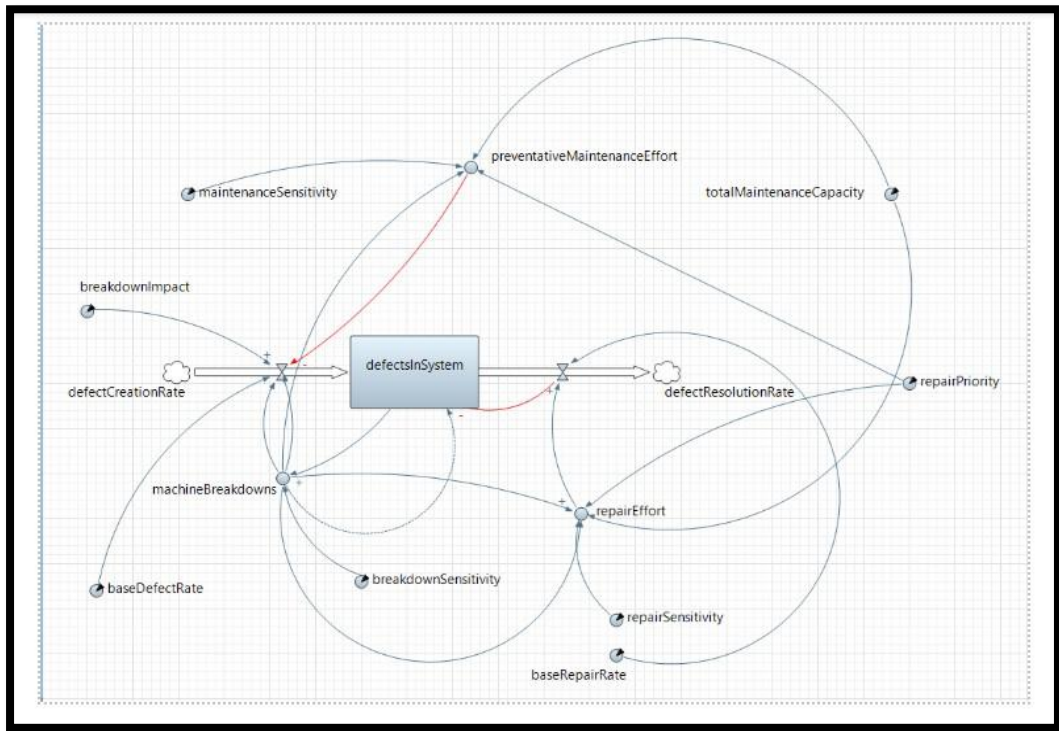
more defects) repair loop (breakdowns → repair effort → defects

resolution) preventative loop (breakdowns → prevention → reduced

new defects)

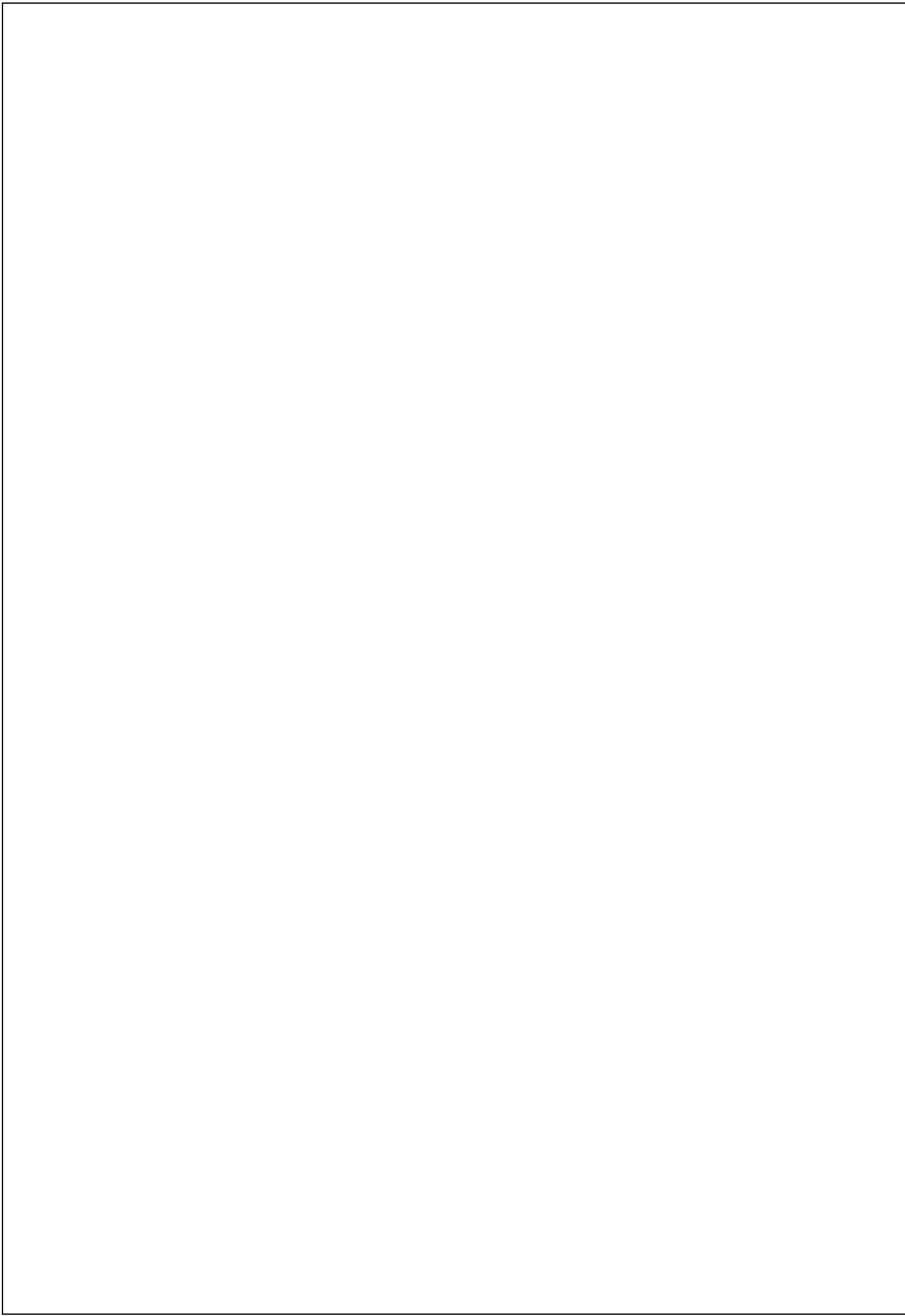
creates the dynamic and realistic representation of how smart management could shift the system from reactive to proactive control.

3e. Consider also that time of the maintenance personnel is limited and can either be spent on repair or preventative maintenance. Explain any assumptions about priority between these two activities and add any additional structure. Discuss loops and overall behaviour.



In this diagram, a realistic production environment is modeled in which maintenance resources are scarce and must be shared between the repair of defects and the prevention of future defects.

The center contains the stock `defectsInSystem` which represents the total number of unresolved defects at any instant in time. Defects enter the system via an inflow, `defectCreationRate`, which is put into effect by `machineBreakdowns`. The size of these breakdowns increases with the rise in defects as controlled by the parameter `breakdownSensitivity`.



There are two main strategies for defect management: repairing defects once they occur and preventing their occurrence. Both `repairEffort` and `preventativeMaintenanceEffort` compete for a common resource, `totalMaintenanceCapacity`. This signifies an important constraint — the factory has only limited time, manpower, or resources at its disposal for maintenance. This trade-off is introduced by the variable `repairPriority` which determines how much of the total maintenance capacity is dedicated to repair:

$$\begin{aligned} \text{repairEffort} &= \text{totalMaintenanceCapacity} \times \text{repairPriority} \\ \text{preventativeMaintenanceEffort} &= \text{totalMaintenanceCapacity} \times (1 - \text{repairPriority}) \end{aligned}$$

The repair effort improves the `defectResolutionRate` (outflow), which decreases the stock of defects. The preventative maintenance effort reduces the `defectCreationRate` by addressing the root causes of breakdowns before they happen, based on `maintenanceSensitivity`.

Now this model comprises: A reinforcing loop where defects create breakdowns, which in turn create further defects. A balancing loop for repair based on the reduction of defects after they arise. A counterbalancing loop for preventing new defect inflows. The resource constraints limit the capacity of both balancing loops and thus make the whole system sensitive to the way in which priorities are assigned.

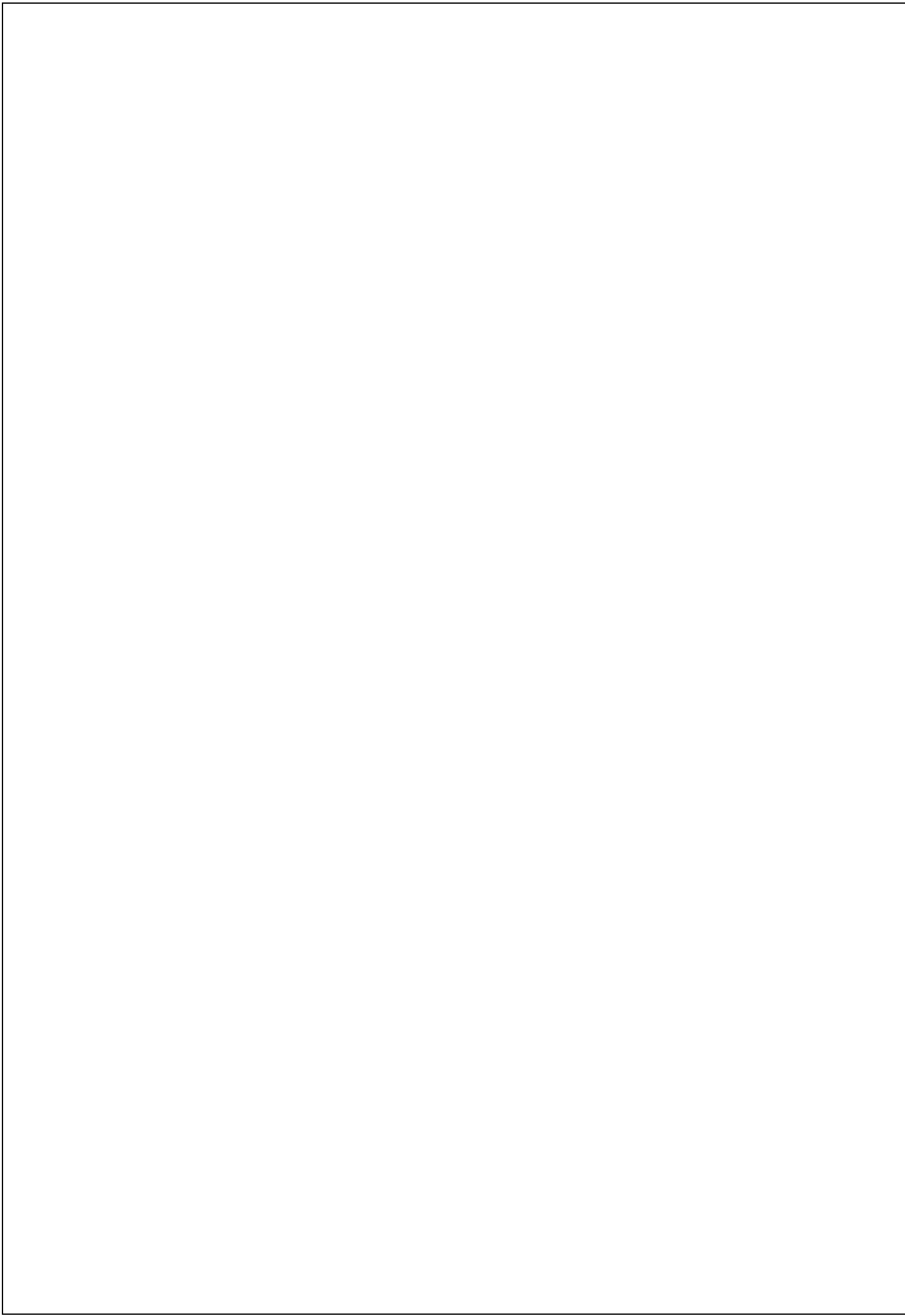
New Elements Introduced in the Model:

To reflect this trade-off, the following components have been added:

totalMaintenanceCapacity: A fixed resource representing the maximum effort (e.g.

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time or manpower) available for both repairs and prevention.



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repairPriority: A value between 0 and 1 that determines what portion of the total

- capacity is allocated to repairs. The rest goes to preventative maintenance. **repairEffort**

and **preventativeMaintenanceEffort** are now calculated based on this

split.

Mathematical Relationships:

$$\text{repairEffort} = \text{totalMaintenanceCapacity} \times \text{repairPriority}$$

$$\text{preventativeMaintenanceEffort} = \text{totalMaintenanceCapacity} \times (1 - \text{repairPriority})$$

This ensures that increasing one type of effort automatically reduces the other.

Impact on System Dynamics:

- If **repairPriority**'s high setting indicates:

More resources committed for the purpose of repairs should be applied to eliminating existing defects.

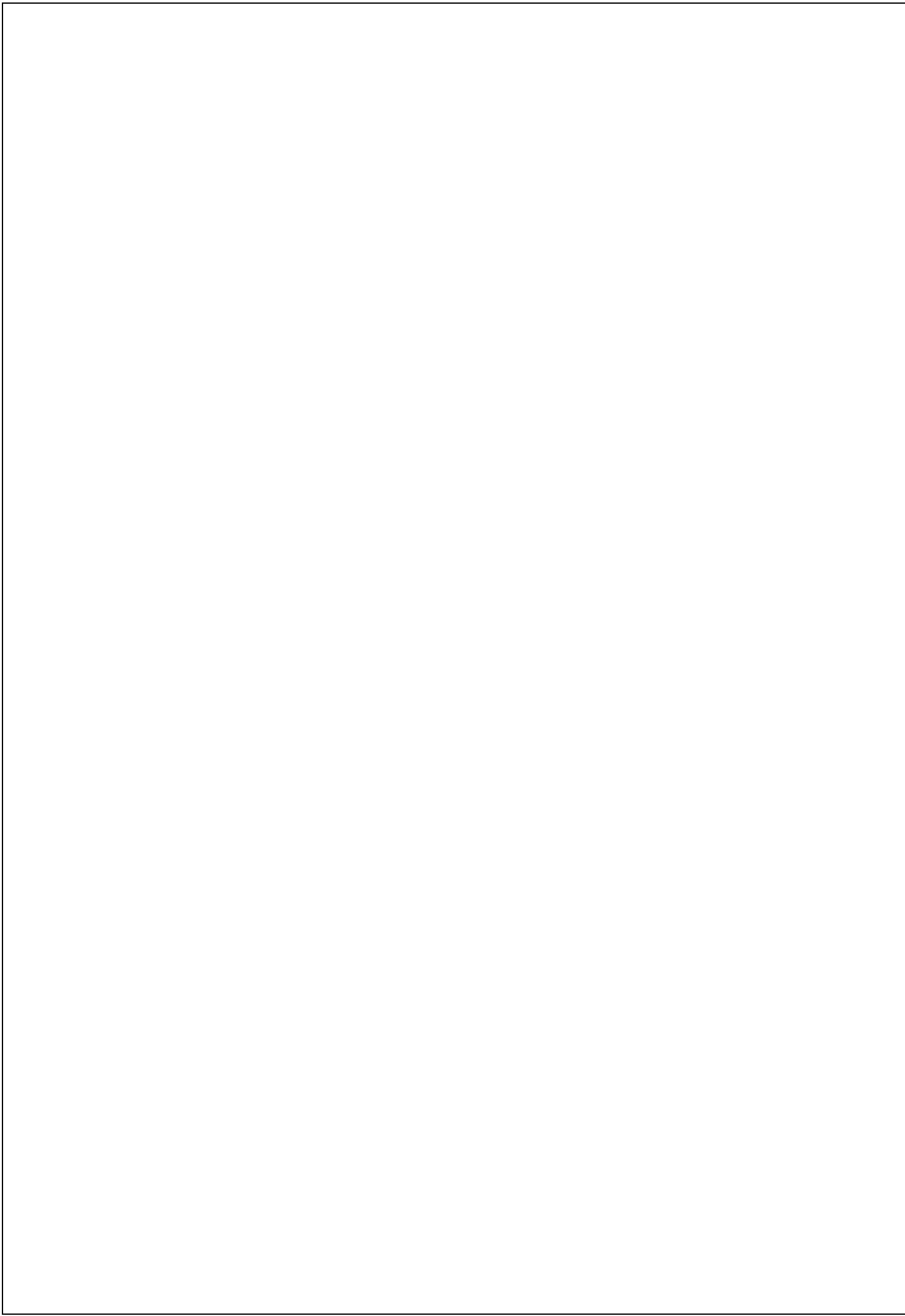
Preventative maintenance suffers, allowing eventually more defects in the future.

- If **repairPriority** is set low:

The system invests more in short term prevention, thereby reducing defect creation.

Existing ones may otherwise stack up in many cases from not being repaired quickly enough.

This trade-off introduces operational tension. If managed poorly, the system could either become overwhelmed by existing issues or experience rising defect creation in the future.



Feedback Loops and Behaviour:

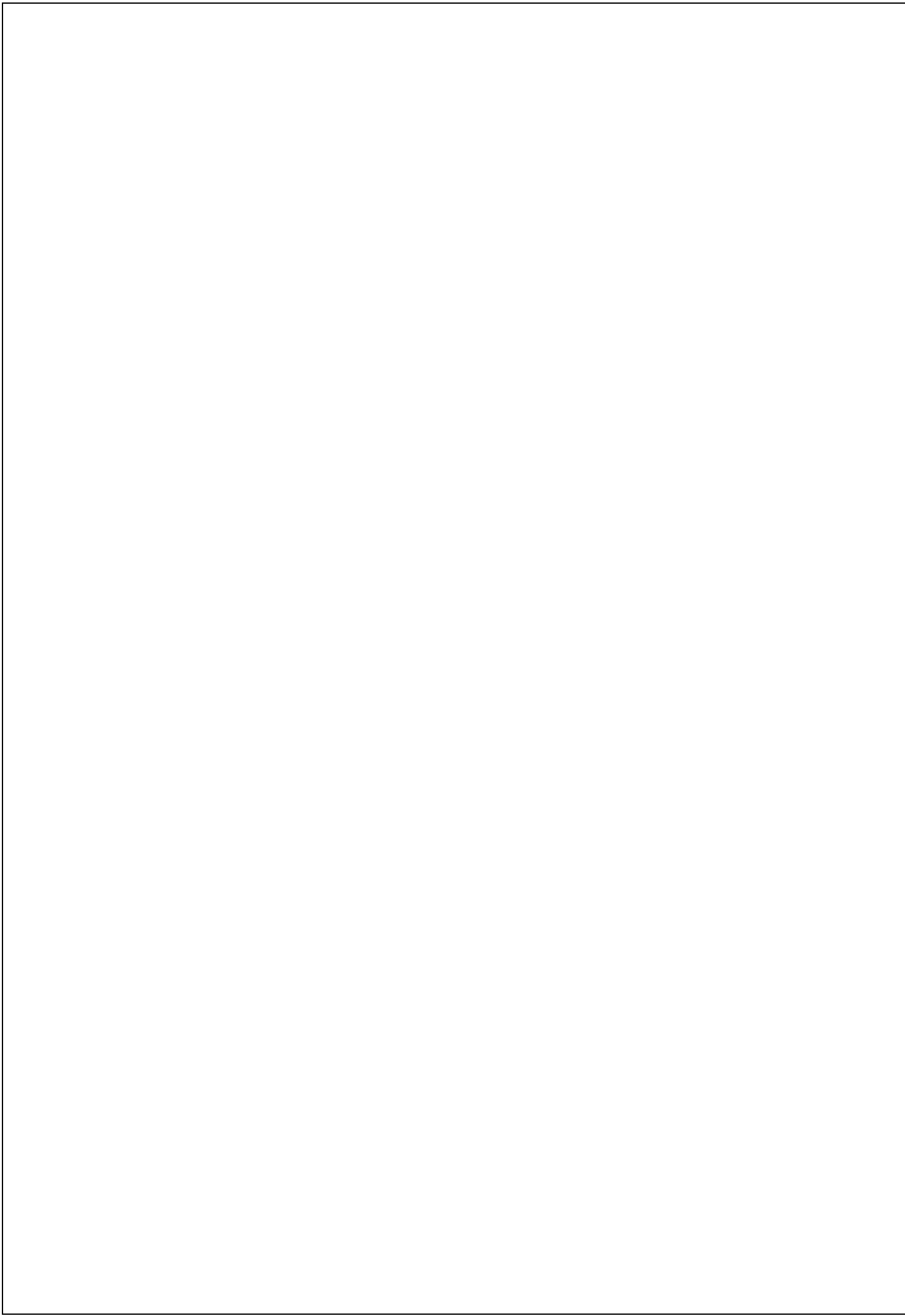
The model now contains **three key loops** and a resource constraint:

Loop Type	Description	Polarity
Reinforcing	Defects → Machine Breakdowns → More Defect Creation → More Defects	Positive
Balancing (Repairs)	More Breakdowns → More Repairs → Higher Defect Resolution → Fewer Defects	Negative
Balancing (Prevention)	More Breakdowns → More Preventative Maintenance → Reduced Defect Creation	Negative
Resource Constraint	Total maintenance effort is fixed and split between repairs and prevention	Constraint

The resource constraint interacts with both balancing loops. It limits their strength, depending on how the available effort is divided.

Expected behaviour over time:

- A well-balanced maintenance capacity, in high value, was able to stabilize the system at low levels of defect and failure.



- Low, poorly allocated capacity (i.e., too much repair work and too little toward prevention) may push the system toward instability.
- Poor decisions (like overemphasizing urgent repairs) can degrade the performance long-term, as excessive preventive measures would mean getting a short glitch in operations. The dynamic arrangement may either have oscillations, chronic instability, or long-term steady state depending on how the trade-off is managed.

Conclusion:

Such a model helps create very realistic and nuanced descriptions of how very resourceconstrained organizations manage competing priorities. Notify that concentrating only on repairs may appear to be a respective answer in the short run, but may allow more defects to come up in the future, as would focusing broadly on prevention that would allow existing problems to grow.

TotalMaintenanceCapacity and repairPriority are sufficient support for strategic thinking and planning. It highlights the importance of dynamic resource allocation and finding the right balance between fixing problems and preventing them — the major challenge in operations management.

This structure allows us to understand some of the trade-offs faced by real operational systems. While excessive focus on repairs will make the system susceptible to imminent future breakdowns, excessive focus on prevention may allow the accumulation of existing defects. The diagram shows that the balance between short-term fixes and longterm stability is critical for defect level management over a given time horizon.

QUESTION: 4

Q4. Mathematical and computational models for describing human behaviour can vary in terms of how transparent they are to different stakeholders. Pick a model from those taught in class that you think is not very transparent, and evaluate it over 5 dimensions. Example dimensions are if decision makers (e.g., lay people) would understand the model, if policy makers would understand it, or if the model can be represented graphically.

Model Selected: Inequity Aversion Model (Fehr & Schmidt, 1999)

The inequity aversion model is a behavioural economic theory that tells why an individual is likely to reject an unfair offer even when it may be against their material interest. The concept extends the expected utility framework to include inequity aversion in the parameters. The theory has been employed mostly in explaining responder behaviour in ultimatum games, where individuals often reject low offers despite receiving nothing.

The utility function proposed by Fehr & Schmidt (1999) is:

$$U_i(x_i, x_j) = x_i - \alpha_i \cdot \max(x_j - x_i, 0) - \beta_i \cdot \max(x_i - x_j, 0)$$

Where:

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xi: the player's own payoff

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- x_j : others' payoffs
- α_i : weight on disadvantageous inequality
- β_i : weight on advantageous inequality

Measures *disadvantageous inequity aversion* (envy) - Dislike of being worse off

Measures *advantageous inequity aversion* (guilt) - Dislike of being better off

1. Understanding by Decision Makers (Lay People)

Transparency: **Low**

Most often, non-expert decision makers such as employees or customers will not understand the mathematical structure of the model. They may understand the words, "I want to be treated fairly," but the functional form with max terms and inequality parameters will not register. It does not match how people naturally think about fairness.

2. Understanding by Policy Makers

Transparency: **Moderate to Low**

The general notion of fairness aversion may have relevance or influence in the thinking of policymakers; however, interpreting the equations of the model in any reasonable way would take training. Variables very often do not communicate intuitive meaning unless subjected to scrupulous explanation and illustration by example. The model offers no straightforward policy guidance unless translated into plain language.

3. Graphical Representation

Transparency: **Moderate**

The inequity aversion model does not have any representation using simple graphs. The piecewise function in this model makes it harder to visualize clearly compared with utility curves in basic economics. Simple bar charts or decision trees can demonstrate behaviours (for example, rejecting unfair offers), yet they cannot represent the deeper underlying structure of the utility formula without a dangerous oversimplification.

4. Parameter Interpretability

Transparency: **Low**

The parameters are abstract. For a non-technical stakeholder, it is unclear what any specific value of or means. Although they encapsulate levels of fairness sensitivity, the model does not give any indication of reasonable thresholds or explanations for interpreting what they mean without prior knowledge.

5. Cognitive Realism

Transparency: **Low to Moderate**

While inequity aversion captures emotional responses such as envy or guilt, it still presumes that people calculate utility by a formula. In real-life decision-making, most people give more weight to rules of thumb or feelings rather than numeric trade-offs. Fast-and-frugal trees or priority heuristics would be more representative models of how human processes about this issue. Accordingly, while the output behaviour may be accurate, the process lacks realism.

Conclusion

The inequity aversion model serves as a useful theoretical tool in explaining behaviour in economic games, particularly rejections of unfair offers. Nevertheless, for most stakeholders, mathematical complexity, non-intuitive parameters, and limited graphical representation afford low transparency. The issue is further worsened through its lack of cognitive realism; hence, these implications are quite difficult to convey to decisionmakers, policymakers, or general audiences without simplification.

Dimensions	Transparency Level	Why?
1. Understanding by Lay People	Low	Abstract parameters
2. Understanding by Policy Makers	Moderate to Low	Core of the idea is clear but model is too complex
3. Graphical Representation	Moderate	Outcomes can be visualized but not formula
4. Interpretability of Parameters	Low	Parameters not meaningful
5. Cognitive Realism	Low to Moderate	People don't think using equations

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